Question 1:

(a) The number (10010001) is basically a signed second complement of the original binary number. Find out its equivalent decimal number. [3m]

(b) Consider the following Sum-of-Products (SOP) Boolean expression:

$$F(A, B, C, D) = \Sigma m(0, 1, 3, 4, 5, 7, 12, 13, 15)$$

i. Using Karnaugh map, simplify the above SOP Boolean expression to provide results in both SOP and Product-of-Sums (POS) forms. [1.5+1.5=3m]

ii. Implement both simplified SOP and POS using logic gates and compute their respective costs based on the following information: [1.5+1.5=3m]

74LS08 Quad Dual Input AND Gates IC:	RM 1.50 per piece
74LS32 Quad Dual Input OR Gates IC:	RM 1.50 per piece
74LS04 Hex Inverters IC:	RM 1.50 per piece

Question 2: A latest car from BMW can be started if one of the conditions are met,

The engine is not in a fault state, and the battery level is sufficient, or The time is within allowed driving hours (6 AM to 11 PM), the engine is not in a fault state, and a valid key fob signal is detected, or The engine is not in a fault state, the battery level is not sufficient, but a valid key insert signal is detected and the time is within allowed driving hours.

Can you advise him to implement this using a logic circuit implemented with

i. AND and OR gates

ii. NAND gates alone

iii. NOR gates alone **[2+2+2= 6m]**

Note: Only answer will be given zero marks. You must explain step by step until you get your answer.